

SUMMARY TABLE: BIOLOGY GRADE 10

Sub-Strand 3.1: Animal Nutrition — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 3.1: Animal Nutrition
Driving Question	DRIVING QUESTION: How do the structures animals use to get food — from an aphid's needle-like mouthparts to a flamingo's filter beak to your own teeth and digestive system — explain how each animal gets the nutrients it needs to survive?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 What Is Animal Nutrition and Why Does It Matter? — Launching the Anchoring Phenomenon	Learners watched video footage of a mosquito piercing skin with its proboscis, a grasshopper chewing maize leaves with mandibles, aphids feeding on plant stems, a flamingo sweeping its filter beak through water at a lake, and a fish eagle tearing a fish with its hooked beak. Learners recorded specific observations in a two-column chart and identified five distinct feeding actions: PIERCE, CHEW, SUCK SAP, FILTER, TEAR.	Learners learned that: (1) animal nutrition is the process by which animals obtain and use food for energy, growth, and repair; (2) different animals possess feeding structures (mouthparts, beaks, teeth) that are adapted to the type of food they eat; (3) the term 'adaptation' describes a structure or behaviour that helps an organism survive — and feeding structures are a key example of structural adaptation; (4) the shape of a feeding structure is directly connected to the type of food it can access (e.g., a needle-like proboscis accesses liquid blood; flat grinding mandibles process solid plant tissue).	Learners began to explain the anchoring phenomenon — why a mosquito can drink blood but a grasshopper cannot — by connecting the STRUCTURE of each animal's feeding apparatus to the FUNCTION it performs and the FOOD it enables access to. Learners constructed a Structure → Function → Food table and drew an annotated starting model of one animal's feeding structure. They posted questions on the Driving Question Board that they cannot yet explain, including questions about what happens INSIDE the animal's body after food enters — the central question of the remaining lessons in this sub-strand.
Lesson 2 Chewing and Biting Mouthparts: How Mandibles Enable Mechanical Breakdown of Solid Food	Ragged, torn holes in maize leaves from the Kakamega farm; a labelled grasshopper mouthpart diagram showing five components including hard, toothed, sideways-moving mandibles; a termite mouthpart diagram showing similar chewing structures adapted to woody material.	Chewing insects possess mandibles that are hardened, toothed, and move sideways (medially) to cut and grind solid food. The labrum, maxillae, labium, and hypopharynx support the mandibles in processing food. This physical tearing and grinding is called mechanical digestion. Both grasshoppers and termites share this mouthpart pattern because they both feed on solid organic material.	The ragged holes in the Kakamega maize leaves are caused by the grasshopper using its mandibles to cut and grind solid leaf tissue — a process of mechanical digestion. The hardness and serrated surface of the mandibles, combined with their sideways movement, make them structurally suited to processing solid plant material. This is an example of structural adaptation: the shape of the mandible

			directly enables the grasshopper to obtain nutrients from a food source that requires physical force to break down. This evidence advances the driving question by showing that one key factor determining the shape of an animal feeding structure is the physical form of its food — solid tissue requires a hard, grinding tool.
Lesson 3 Piercing and Sucking Mouthparts: How Does a Mosquito Drink Blood but a Grasshopper Cannot?	Learners watched video clips showing aphids inserting their stylets into maize stems in a Kakamega-style farm setting and a mosquito proboscis penetrating skin in slow motion, recording structural features and feeding behaviours on a structured comparison chart. They also revisited their Lesson 1 predictions about the mosquito-vs-grasshopper phenomenon.	Learners constructed physical models of piercing-sucking mouthparts using straws and foam, labelling the labium, food canal, salivary canal, and stylet bundle. They wrote CER explanations linking the needle-like stylet structure to liquid-food access and the absence of visible chewing damage. They updated the class Driving Question Board with new evidence about aphids (Kakamega maize) and mosquitoes (Nairobi), and extended their cumulative mouthpart diversity model to contrast piercing-sucking with the chewing mouthparts from Lesson 2.	Piercing-sucking mouthparts — found in mosquitoes, aphids, and tsetse flies — consist of a multi-channel stylet bundle enclosed in a protective labium sheath. The stylet penetrates plant phloem tissue or animal skin through a gradual rocking motion, without tearing the surface. Saliva is injected through the salivary canal to prevent clotting or aid sap flow, while liquid food (blood or phloem sap) is drawn up through the food canal. This structural specialisation allows these insects to access nutrient-rich liquid food sources unavailable to chewing insects — explaining why an aphid can wilt a maize plant without leaving a single hole, and why a mosquito can draw blood painlessly at first, connecting directly to the anchoring phenomenon.
Lesson 4 Crushing and Tearing Bird Beaks: How Does a Fish Eagle Eat What a Flamingo Cannot?	Photographs and labelled diagrams of the fish eagle, white-backed vulture, African grey hornbill, and pied crow beaks, showing differences in tip sharpness, curvature, gape width, and serration of the tomia (inner beak edge)	Beak morphology in Kenyan birds directly reflects diet: the fish eagle has a sharply hooked, pointed beak for gripping and tearing fish flesh; the vulture has a hooked beak with wide gape for reaching into carcasses; the hornbill has a large, blunter curved beak for crushing and tossing hard fruits and invertebrates; all three represent adaptations in which structure determines the food an animal can access	Using CER statements, learners explained that the specific structural features of each beak — curvature, tip sharpness, gape width — enable specific feeding actions (tearing, gripping, crushing), and that these adaptations are as functionally necessary as an aphid stylet or a grasshopper mandible, advancing the driving question evidence that feeding structures across all animals reflect the food source the animal relies on for nutrients
Lesson 5 Probing and Filtering Bird Beaks: How Beak Morphology Reveals Diet and Habitat	Learners watched flamingos at Lake Nakuru filter-feeding with their bent, lamellae-lined beaks and sunbirds probing Kenyan flowers with long, curved beaks. They recorded specific structural features — including the upside-down beak position, tongue pumping action, and comb-like	Learners learned that beak morphology is a direct structural adaptation to food source and habitat: the flamingo's beak acts as a biological sieve, using lamellae to filter algae and small crustaceans from alkaline Lake Nakuru water, while the sunbird's curved beak and tubular tongue allow it to	Learners explained that across all animals studied in Lessons 1–5 — from the grasshopper's mandibles and the mosquito's fascicle to the flamingo's filter beak and the sunbird's probing beak — the pattern is the same: every feeding structure is shaped precisely to match the food

	lamellae of the flamingo, and the decurved beak and tubular tongue of the sunbird — on structured observation charts.	reach nectar deep inside Kenyan flowers. They applied the Claim-Evidence-Reasoning framework to explain why each beak cannot perform the other's function.	source and the habitat in which the animal finds that food. This principle of structure-determines-function-determines-diet was added to the cumulative class model and the Driving Question Board as the class's emerging explanation, setting up the shift in Lessons 6–10 toward what happens to food inside the animal's body.
Lesson 6 Dentition in Mammals: Tooth Shape, Diet, and the Structure-Function Connection	Skull diagrams and Maasai Mara wildlife photographs of a lion, cow, and human; a 3-minute multimedia clip of a lion feeding on a wildebeest and a wildebeest chewing cud at the Maasai Mara; tooth counts recorded in a structured evidence table.	The four types of mammalian teeth (incisors, canines, premolars, molars) differ in shape and function; dental formulae can be calculated from skull diagrams; carnivores (lion) have prominent canines and carnassial teeth for gripping and shearing meat; herbivores (cow) lack upper incisors and have broad, ridged molars for grinding grass; omnivores (humans) have a balanced set of all four tooth types reflecting a mixed diet.	Tooth shape is a structural adaptation directly linked to diet: the lion's elongated canines and shearing carnassials allow it to grip and slice muscle tissue at the Maasai Mara, while the cow's dental pad and wide molars allow it to crop and grind tough grass. This structure-function relationship in mammalian dentition extends the pattern seen across all animals in this unit — from the mosquito's piercing stylet to the flamingo's filtering lamellae — confirming that the shape of a feeding structure determines what food an animal can access and how it begins to extract nutrients.
Lesson 7 The Human Digestive System: Tracing a Meal of Ugali and Sukuma Wiki	Learners built a physical model of the human digestive system using rubber tubing, balloons, a funnel, rope, and a nylon stocking to represent the oesophagus, stomach, small intestine, large intestine, and rectum. They moved a clay food bolus through the model, narrating the journey of a meal of ugali and sukuma wiki from mouth to anus, and labelled where mechanical digestion, chemical digestion, and absorption of starch, vitamins, iron, and water occur.	The human digestive system is a series of specialised organs — mouth, oesophagus, stomach, small intestine, large intestine, and rectum — each structurally adapted to perform a specific function. Starch in ugali is broken down by amylase in the mouth and pancreatic juice in the small intestine, producing glucose that is absorbed through the villi into the bloodstream. Vitamins and iron from sukuma wiki are also absorbed in the small intestine. Water is reclaimed in the large intestine. Structure determines function at every level — from the length of the small intestine (maximising absorption surface area) to the muscular walls of the stomach (enabling mechanical churning).	Just as the mosquito's proboscis, the grasshopper's mandibles, and the flamingo's lamellae are external structures adapted to obtain specific foods, the human digestive system is a set of internal structures adapted to process and extract nutrients from a varied diet. The mosquito bypasses the need for a complex digestive system because blood is already nutrient-rich and dissolved — but a human eating ugali and sukuma wiki needs a multi-organ system to chemically break down complex carbohydrates, proteins, and fibre before nutrients can enter the blood. Structure determines function — from beak to bowel.
Lesson 8 Enzymes in Digestion: How Ugali and Sukuma Wiki Are Chemically Broken Down	When I chewed plain ugali for two minutes, the taste changed from starchy to slightly sweet. When I tested unchewed ugali with iodine it turned blue-black, and with Benedict's reagent it remained blue. When I tested chewed ugali with iodine it turned yellow-brown, and with Benedict's reagent it	Amylase in saliva chemically breaks down starch (a large, insoluble carbohydrate) into reducing sugars (small, soluble molecules) in the mouth and small intestine. Pepsin in the stomach breaks down proteins such as those in nyama choma. Lipase in the small intestine breaks down fats from foods like	The sweet taste I noticed when chewing ugali was caused by amylase converting starch molecules into sugar molecules. The colour changes in the food tests provided evidence of this molecular transformation — iodine could no longer find starch to react with, and Benedict's reagent found reducing

	turned brick-red after heating.	avocado into fatty acids and glycerol. Enzymes are biological catalysts — they are specific to their substrate and are not used up in the reaction.	sugar that was not present before. This explains how the organs of the digestive system, which I labelled in Lesson 7, actually carry out their function at the molecular level. It also connects back to the anchoring phenomenon: a mosquito uses salivary enzymes to prevent blood clotting just as I use amylase to begin digesting ugali — both are molecular adaptations matched to a specific food source.
Lesson 9 Absorption, Assimilation, and Malnutrition: From Villi to Village Health	We examined a detailed diagram of the ileum wall and identified villi and microvilli as finger-like projections that massively increase the surface area available for absorption. We observed two distinct pathways inside the villus: capillaries that receive glucose and amino acids, and a central lacteal that receives fatty acids and glycerol. We also read real Kenyan health data showing that children in Kilifi County can develop kwashiorkor from protein-deficient diets and that many adolescent girls in Nairobi have iron-deficiency anaemia.	Digested nutrients are absorbed in the ileum across the thin epithelial cells of the villi into the blood or lymph. Glucose and amino acids diffuse into blood capillaries and travel to the liver via the hepatic portal vein. Fatty acids and glycerol recombine into fat droplets that enter the central lacteal and travel through the lymphatic system. Assimilation means body cells actually use those absorbed nutrients — glucose in respiration, amino acids to build proteins. Malnutrition occurs when the diet lacks adequate nutrients, and even healthy villi cannot absorb what is not consumed.	The entire journey from a meal of ugali and sukuma wiki — chewed in the mouth, digested by enzymes in the stomach and small intestine — ends with small soluble molecules crossing the villus wall into the blood. The structural features of villi (thin walls, large surface area, rich blood supply, central lacteal) make this transfer efficient. When any part of this system fails — whether because of an inadequate diet or damaged villi — cells are deprived of nutrients they need, leading to conditions like kwashiorkor or anaemia. This connects back to the anchoring phenomenon: a mosquito drinking liquid blood delivers ready-made amino acids directly to its gut for absorption, bypassing the need for extensive mechanical digestion — a shortcut that a chewing grasshopper with solid plant food cannot take.
Lesson 10 Synthesis and Model Finalisation: From Aphid Stylets to Human Villi — One Purpose, Many Structures	Learners observed their own Lesson 1 initial models alongside their finalised Lesson 10 cumulative models during the Gallery Walk, noting structural differences in how the anchoring phenomenon animals (grasshopper, aphid, mosquito, flamingo, fish eagle) were represented. They also observed peer models during the Gallery Walk, using sticky-note feedback to identify evidence-linked strengths and gaps. During the Popcorn Round, learners observed how individual contributions built toward a collective scientific explanation of the driving question.	Learners consolidated the unit's core understanding: that all animal feeding structures — from the aphid's needle-like stylet that pierces phloem vessels in maize stems in Kakamega, to the flamingo's comb-like lamellae that filter cyanobacteria from Lake Nakuru's alkaline water, to the human small intestine's villi that maximise glucose and amino acid absorption — are structural adaptations that serve the same ultimate biological purpose: obtaining nutrients for survival. Learners also learned to evaluate Kenyan food choices (githeri, sukuma wiki, ugali, mandazi, nyama choma) using nutrient density relative to cost, connecting digestion and absorption	Learners constructed individual Final Explanations using the CER+Evidence framework (Claim–Evidence–Reasoning with four cross-unit evidence references), answering the driving question: 'How do the structures animals use to get food explain how each animal gets the nutrients it needs to survive?' The class collectively completed the Driving Question Board, closing all outstanding inquiry questions and co-authoring a 'Class Final Explanation.' Learners explained how the shape of a feeding structure is inseparable from its function — a grasshopper's grinding mandibles can process tough maize leaf cellulose but cannot pierce skin; a

		biology to real-world dietary decisions.	mosquito's fascicle-bundled proboscis can navigate dermis and locate capillaries but cannot chew solid tissue. Structure constrains and enables function, and function determines what nutrients are accessible — this principle, applied from beak tip to intestinal microvillus, is the unifying explanation of Sub-Strand 3.1.
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ATTRIBUTION AND LICENSING

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